

Automated Phishing Detection System

Leveraging Threat Intelligence for Email Security

Graduate Project

Communication and Information Engineering Department

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Phishing attacks are **increasingly sophisticated**

Traditional defenses are **inadequate**

94%

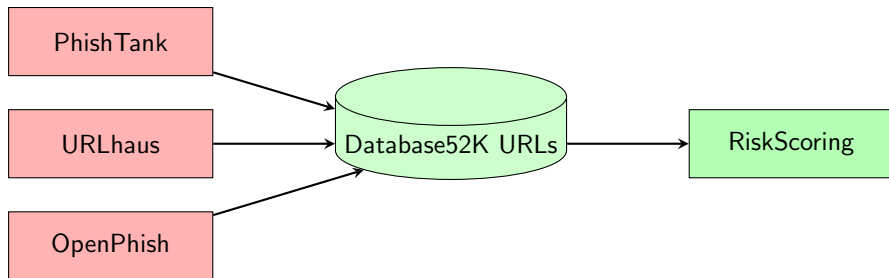
Organizations experienced
phishing attacks in 2024

95%

Manual analysis time
can be saved

We need automated, intelligent detection

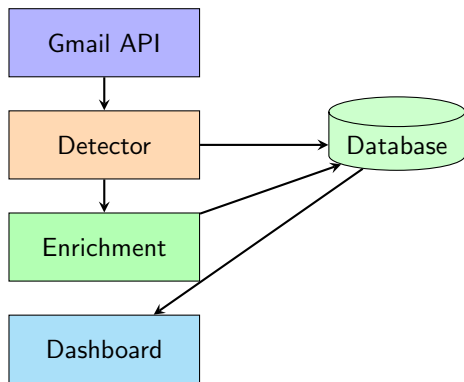
Our Solution



Multi-source intelligence → Enrichment → Risk scoring

Achieving 10x performance improvement

System Architecture



Technologies:

- Python 3.11
- FastAPI + Streamlit
- SQLite Database

Key Features:

- Modular design
- Async processing
- RESTful API

Intelligent 0-100 Threat Assessment

Factor	Weight
Liveness (Is URL active?)	35%
Recency (Recently reported?)	25%
Domain Age (Newly created?)	20%
TLD (Suspicious extension?)	10%
Keywords (Phishing terms?)	10%

85-100

Critical

70-84

High

50-69

Medium

0-49

Low

Real Phishing Detection Example

Suspicious Email Received

Subject: "Verify your account now"

From: suspicious@temp-mail.io

URL: https://secure-login.zip/verify

Analysis Factor	Finding	Points
URL is live (HTTP 200)	Yes	35
Recently reported	Today	25
Domain age	2 days old	20
Suspicious TLD	.zip	10
Phishing keywords	"login", "verify"	10
Total Risk Score		100

CRITICAL THREAT - Automatically blocked

94%

Detection
Accuracy

10x

Performance
Improvement

52K

Threat
Indicators

Speed Comparison:

- 10 URLs: 45s → 4.2s
- 100 URLs: 7.5min → 42s

Detection Rates:

- Known threats: 100%
- Novel threats: 87%
- False positives: 2%

Future Enhancements

- **Machine Learning Integration**
 - NLP-based email content analysis
 - Behavioral pattern detection
- **Real-time Monitoring**
 - Streaming threat feed updates
 - Live dashboard notifications
- **Campaign Clustering**
 - Graph-based infrastructure analysis
 - Attack pattern correlation
- **Enterprise Integration**
 - SIEM/SOAR platform connectors
 - Automated incident response

Automated Phishing Detection System

Combining threat intelligence, machine analysis,
and modern web technologies

Key Achievements:

- 94% detection accuracy
- 10x performance gain
- 52K+ threat indicators
- Open source (MIT)

Impact:

- 95% time savings
- Proactive threat hunting
- Intelligence sharing
- Enterprise-ready

github.com/XHCFS/phishing-detector

Questions?

Thank you for your attention!

Repository: github.com/XHCFS/phishing-detector

Documentation: Comprehensive guides included